Lecture-02

* Rational Function:

A rational function is a function that is the ratio of polynomials. $f(x)$ will be a rational function if we can write it as following way,

$$f(x) = \frac{P(x)}{Q(x)}$$

where $P(x)$ and $Q(x)$ are polynomials such that $Q(x) \neq 0$.

Ex. $f(x) = \dfrac{x^2+x-2}{x^3+3x+8}$;

N.B: Numerator of the rational functions can be constants but denominators must not be constants.

Ex. $f(x) = \dfrac{2}{5x^3+3x}$ is rational function

but $f(x) = \dfrac{x^2+8x+3}{9}$ is not a rational function

* Limits of Rational Functions as $x \to \pm\infty$:

One technique for determining the end behavior of a rational function is to divide each term in the numerator and denominator by the highest power of x that occurs in the denominator.

*** Example-01:** Find $\lim_{x \to \infty} \frac{3x+5}{6x-8}$

Ans: $\lim_{x \to \infty} \frac{3x+5}{6x-8} = \lim_{x \to \infty} \frac{x\left(3+\frac{5}{x}\right)}{x\left(6-\frac{8}{x}\right)}$

$= \lim_{x \to \infty} \frac{3+\frac{5}{x}}{6-\frac{8}{x}} = \frac{3+\frac{5}{\infty}}{6-\frac{8}{\infty}} = \frac{3+0}{6-0}$

$= \frac{1}{2}$.

*** Homework:** Find

a) $\lim_{x \to -\infty} \frac{4x^2-x}{2x^3-5}$; b) $\lim_{x \to \infty} \frac{5x^3-2x^2+1}{1-3x}$

Ans: a) 0 ; b) $-\infty$;

* Example-02: Find $\lim_{x \to +\infty} \frac{\sqrt{x^2+2}}{3x-6}$

(solution)

Let us find out $\lim_{x \to +\infty} \frac{\sqrt{x^2+2}}{3x-6}$

It would be helpful to manipulate the function so that the powers of x are transformed to powers of $\frac{1}{x}$. This can be achieved by dividing the numerator and denominator by $|x|$ and using the fact that $|x| = \sqrt{x^2}$.

$$\lim_{x \to \infty} \frac{\sqrt{x^2+2}}{3x-6} = \lim_{x \to \infty} \frac{\frac{\sqrt{x^2+2}}{|x|}}{\frac{3x-6}{|x|}}$$

$$= \lim_{x \to \infty} \frac{\frac{\sqrt{x^2+2}}{\sqrt{x^2}}}{\frac{3x-6}{x}}$$

$$= \lim_{x \to \infty} \frac{\sqrt{\frac{x^2+2}{x^2}}}{\frac{x(3-6/x)}{x}}$$

$$= \lim_{x \to \infty} \frac{\sqrt{1+2/x^2}}{3-6/x}$$

$$= \frac{\sqrt{\lim_{x\to\infty} 1 + \lim_{x\to\infty} \frac{2}{x^2}}}{\lim_{x\to\infty} 3 - \lim_{x\to\infty} \frac{6}{x}}$$

$$= \frac{\sqrt{1 + 2/\infty}}{3 - \frac{6}{\infty}}$$

$$= \frac{\sqrt{1+0}}{3} = \frac{1}{3} \text{ Ans.}$$

* Indeterminate forms:

An indeterminate form is a mathematical expression whose value cannot be determined. some indeterminate forms are of following type: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, 0^0, ∞^0, 1^∞

* Applying L' Hôpital's Rule: for calculating the limits of indeterminate form

1. check that the limit of $\frac{f(x)}{g(x)}$ is an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

2. Differentiate f and g separately.

3. Find the limit of $\frac{f'(x)}{g'(x)}$. If the limit is finite, $+\infty$ or $-\infty$, then it is equal to the limit of $\frac{f(x)}{g(x)}$.

★ Example-03: Find $\lim_{x \to 0} \frac{\sin x}{x}$.

Ans: $\lim_{x \to 0} \frac{\sin x}{x}$ $\left[\frac{0}{0} \text{ if } x \to 0\right]$

$= \lim_{x \to 0} \frac{\cos x}{1}$; [differentiating Both Numerator & Denominator with respect to x]

$= \frac{\cos 0}{1} = 1$

★ similarly $\lim_{x \to 0} \frac{\tan x}{x} = 1$

★ Example-04: Find $\lim_{x \to +\infty} \left(\frac{x}{e^x}\right)$

Ans: $\lim_{x \to +\infty} \frac{x}{e^x}$; $\left[\frac{\infty}{\infty}\right]$

$= \lim_{x \to +\infty} \frac{1}{e^x}$;

$$= \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$$

* Determining limits when the indeterminate forms are like $0^0, \infty^0, 1^\infty$:

Indeterminate forms of these type can be sometimes be evaluated by first introducing a dependent variable.

$$y = [f(x)]^{g(x)}$$

$$\Rightarrow \ln y = g(x) \ln[f(x)]$$

* Example-05: Find $\lim_{x \to 0} (1+\sin x)^{1/x}$

Solution

let, $y = (1+\sin x)^{1/x}$

$$\Rightarrow \ln y = \ln[(1+\sin x)^{1/x}]$$

$$\Rightarrow \lim_{x \to 0} \ln y = \lim_{x \to 0} \frac{1}{x} \ln(1+\sin x)$$

$$\ln(1+\sin x)$$

Here, $\lim_{x \to 0} \frac{\ln(1+\sin x)}{x}$; $\left[\frac{0}{0}\right]$

$$= \lim_{x \to 0} \frac{\left(\frac{\cos x}{1+\sin x}\right)}{1}$$

$$= \frac{\cos 0}{1+\sin 0} = \frac{1}{1} = 1$$

Therefore, $\lim_{x \to 0} \ln y = 1$

$$\Rightarrow \lim_{x \to 0} e^{\ln y} = e^1$$

$$\Rightarrow \lim_{x \to 0} y = 1$$

$$\therefore \lim_{x \to 0} (1+\sin x)^{1/x} = e$$

Ans.

* Example-07: Find $\lim_{x \to \infty} \left(1+\frac{2}{7x}\right)^{3x+5}$

* continuity:

A function f is said to be continuous at $x=c$ if the following conditions are satisfied :—

1) $f(c)$ is defined.

2) $\lim_{x \to c} f(x)$ exists.

3) $\lim_{x \to c} f(x) = f(c)$.

* **Example:01:** Determine whether the following function is continuous at $x=3$. $f(x) = \begin{cases} x+2 ; & -1 \le x < 3 \\ 14-x^2 ; & 3 \le x < 5 \end{cases}$

Ans:

Given that $f(x) = \begin{cases} x+2 ; & -1 \le x < 3 \\ 14-x^2 ; & 3 \le x < 5 \end{cases}$

The value of the function at $x=3$ is, $f(3) = 14 - 3^2 = 5$

Now, let see whether $\lim_{x \to 3} f(x)$ exists or not.

As x approaches 3 from the left. i-e, L.H.L ~~at~~ of $f(x)$ at $x=3$ is

$$L.H.L = \lim_{x \to 3^-} f(x)$$

$$= \lim_{x \to 3^-} x+2 = 3+2 = 5$$

As x approaches 3 from the right

$$R.H.L = \lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} (14-x^2) = 14-3^2 = 5$$

since L.H.L = R.H.L at $x=3$, implies that $\lim_{x \to 3} f(x)$ exists and

$$\lim_{x \to 3} f(x) = 5$$

Here we observe that,

$$\lim_{x \to 3} f(x) = f(3) = 5$$

since the function $f(x)$ satisfies all the three conditions of continuity therefore, the function $f(x)$ is continuous at $x=3$.

Ans.

* Do yourself: let $g(x) = \begin{cases} 5-x ; & -1 \le x \le 2 \\ x^2-1 ; & 2 < x < 3 \end{cases}$

Determine whether the function $g(x)$ is continuous at $x=2$

* Definition:

A function f is said to be continuous on a closed interval $[a,b]$ if the following conditions are satisfied:

1) f is continuous on the open interval (a,b).

2) f is continuous from the right at a i.e, $\lim_{x \to a^+} f(x) = f(a)$

3) f is continuous from the left at b i.e,

$$\lim_{x \to b^-} f(x) = f(b)$$

* <u>Example</u>: A function is defined as $f(x) = \sqrt{x-2}$. Is the function continuous on the interval $[2, \infty)$.

$\boxed{\text{solution}}$

Given that, $f(x) = \sqrt{x-2}$

1) Let us consider the open interval $(2, \infty)$. Let c be any point on $(2, \infty)$. i.e, $c \in (2, \infty)$

The value of the function $f(x)$ at $x = c$, $f(c) = \sqrt{c-2}$

Here, $\lim_{x \to c} f(x) = \lim_{x \to c} \sqrt{x-2} = \sqrt{c-2}$

for any point $c \in (2, \infty)$ the function is defined and is continuous. It is true for all points $c \in (2, \infty)$, Therefore the function $f(x)$ is continuous on the open interval $(2, \infty)$.

2) As x approaches 2 from the right,

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} \sqrt{x-2} = \sqrt{2-2} = 0$$

and at $x = 2$, $f(2) = 0$

Therefore, $\lim_{x \to 2^+} f(x) = f(2)$. ~~It~~

It means that the function is continuous from the right at $x = 2$.

3) As x approaches $+\infty$ from the left, there will be no point for which the function is not

continuous and the value of the function at the end point will approach $+\infty$.

So, we can say that, the function $f(x)$ satisfies all the three conditions of continuity.

Therefore, the function is continuous on the interval $[2, \infty)$

* <u>Do yourself</u>:

What can you say ~~that~~ about the continuity of the function $f(x) = \sqrt{9-x^2}$ on the closed interval $[-3, 3]$?